作业纸

班级：未央-水木22　姓名：郁然，　学号：2022012835　科目：数值分析　第 1 页

1. (1) $\|f\|_\infty = \max\limits_{0\le x\le 1}|f(x)| = \dfrac{m^m n^n}{(m+n)^{m+n}}$

$\|f\|_1 = \int_0^1 |f(x)|\,dx = \dfrac{\sum\limits_{k=0}^{n} C_n^k (-1)^k}{k+m+1} \sum\limits_{k=0}^{n} \dfrac{C_n^k (-1)^k}{k+m+1}$

$\|f\|_2 = \left(\int_0^1 x^{2m}(1-x)^{2n}dx\right)^{\frac{1}{2}} = \sqrt{\sum\limits_{k=0}^{2n} \dfrac{C_{2n}^k (-1)^k}{k+2m+1}}$

(2) $\|f\|_\infty = \max\limits_{0\le x\le 1}|f(x)| = \dfrac{2^{10}}{e}$

$\|f\|_1 = \int_0^1 f(x)\,dx = 9864101 - \dfrac{26813184}{e}$

$\|f\|_2 = \left(\int_0^1 f(x)^2 dx\right)^{\frac{1}{2}} = \sqrt{\dfrac{685762383331999}{8} - \dfrac{50671367112320g}{8e^2}}$

2. (1) 令 $f(x)\equiv C$

则 $\int_a^b f'(x)\cdot f'(x)\,dx = 0$　矛盾

(2) $(f,g)=(g,f)$ 显然

$\forall h\in C^1[a,b]$, $\forall \alpha,\beta\in\mathbb{C}$

$(\alpha f+\beta g, h) = \int_a^b (\alpha f'+\beta g')h'dx + (\alpha f+\beta g)h$

$= \alpha(f,h)+\beta(g,h)$

$(f,f) = \int_a^b [f'(x)]^2 dx + [f(x)]^2$

$\int_a^b [f'(x)]^2 dx \ge 0$　$[f(x)]^2 \ge 0$

$\therefore (f,f)\ge 0$　~~若 $[f(x)]^2=0$ 时，$f(x)$~~

当 $(f,f)=0$，此时 $[f(x)]^2=0 \Leftrightarrow f(x)=0$，此时 $\int_a^b [f'(x)]^2dx=0$ 成立

~~构成内积~~

3. $\psi_1(x) = x + a_{10}$　$\varphi_0(x) = \sqrt{\frac{1}{\pi}}$

$a_{10} = -\int_{-1}^1 \rho(x)\,x\,dx = 0$　$\psi_1(x)=x$

$\varphi_1(x) = \dfrac{\psi_1(x)}{\|\psi_1(x)\|_2} = \sqrt{\dfrac{3}{\pi}}x$　$P_1(x) = \sqrt{\dfrac{3}{\pi}}x$

$\psi_2(x) = x^2 + a_{21}\varphi_1(x) + a_{20}\varphi_0(x)$

$(\psi_2,\varphi_1)=0$　$(\psi_2,\varphi_0)=0$ $\Rightarrow$ $a_{20} = -\frac{1}{2}$　$a_{21}=0$　$\psi_2(x) = x^2-\frac{1}{2}$

$P_2(x) = \sqrt{\dfrac{5}{\pi}}\left(x^2-\frac{1}{2}\right)$

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4. (1) $S_n^*(x) = a_1\varphi_1(x) + a_0\varphi_0(x)$

$\varphi_1(x) = x$

$\varphi_0(x) = 1$

$$\alpha_1 = \frac{(f, \varphi_1(x))}{(\varphi_1(x), \varphi_1(x))}$$

$$\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{pmatrix}\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} \ln 3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 \\ 4 & \frac{26}{3} \end{pmatrix}\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} \ln 3 \\ 2 \end{pmatrix}$$

$\Rightarrow S_n^*(x) = 1.1410 - 0.2958x$

(2) $S_n^*(x) = a_1\varphi_1(x) + a_0\varphi_0(x)$

$$\begin{pmatrix} 1 & \frac{3}{2} \\ \frac{3}{2} & \frac{7}{3} \end{pmatrix}\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 2\ln 2 - 1 \\ 2\ln 2 - \frac{3}{4} \end{pmatrix}$$

$S_n^*(x) = -0.6374 + 0.6825x$

5. $$\begin{pmatrix} (\varphi_0, \varphi_0) & (\varphi_0, \varphi_2) \\ (\varphi_0, \varphi_2) & (\varphi_2, \varphi_2) \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} (\varphi_0, y) \\ (\varphi_2, y) \end{pmatrix}$$

$a = 0.972578569$

$b = 0.05003512 4$

$y = 0.972578569 + 0.05003512 4x^2$

$\sqrt{\sum_{j=0}(y_j - y(x_j))^2} = 0.122569 2$

6. $b_0 = 1$　$c_0 = 0$　$c_1 = 1$　$c_2 = -\frac{1}{2}$　$c_3 = \frac{1}{3}$　$c_4 = -\frac{1}{4}$　$c_5 = \frac{1}{5}$　$c_6 = -\frac{1}{6}$

$m = 3$　$n = 3$

$$\begin{pmatrix} c_3 & c_4 & c_5 \\ c_4 & c_5 & c_6 \\ c_5 & c_6 & c_7 \end{pmatrix}\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = -\begin{pmatrix} c_4 \\ c_5 \\ c_6 \end{pmatrix} \Rightarrow \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} \frac{207}{304} \\ -\frac{9}{190} \\ \frac{17}{304} \end{pmatrix}$$

$a_0 = c_0 = 0$

$a_1 = c_1 = 1$

$a_2 = c_2 + c_1 b_1 = \frac{1}{304} \frac{55}{304}$

$a_3 = c_3 + c_1 b_2 + c_2 b_1 = -\frac{497}{9120}$

$$R(3,3) = \frac{x + \frac{55}{304}x^2 - \frac{497}{9120}x^3}{1 + \frac{207}{304}x^2 - \frac{9}{190}x^2 + \frac{17}{304}x^3}$$

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记 $\frac{2\pi j}{n} = x_j \quad j=0,1,\cdots,n-1$

$$e^{-\frac{i2\pi jk}{n}} = \delta_{jk} \quad \begin{matrix} j=0,1,\cdots,n-1 \\ k=0,1,\cdots,n-1 \end{matrix}$$

$$\begin{pmatrix} f(x_0) \\ \vdots \\ f(x_{n-1}) \end{pmatrix} = \frac{1}{n} \begin{pmatrix} \delta_{00},\cdots,\delta_{0,n-1} \\ \vdots \qquad\quad \vdots \\ \delta_{n-1,0},\cdots,\delta_{n-1,n-1} \end{pmatrix} \begin{pmatrix} g(x_0) \\ \vdots \\ g(x_{n-1}) \end{pmatrix}$$

$\delta_{ij} = \delta_{ji}$

$$\begin{pmatrix} \delta_{00} & \cdots & \delta_{0,n-1} \\ \vdots & & \vdots \\ \delta_{n-1,0} & \cdots & \delta_{n-1,n-1} \end{pmatrix} \begin{pmatrix} \frac{1}{\delta_{00}} & \cdots & \frac{1}{\delta_{0,n-1}} \\ \vdots & & \vdots \\ \frac{1}{\delta_{n-1,0}} & \cdots & \frac{1}{\delta_{n-1,n-1}} \end{pmatrix} = \begin{pmatrix} n & \cdots & n \\ \vdots & & \vdots \\ n & \cdots & n \end{pmatrix}$$

$$\therefore g\left(\frac{2\pi j}{n}\right) = \sum_{k=0}^{n-1} f\left(\frac{2\pi k}{n}\right) e^{\frac{i2\pi jk}{n}} \quad j=0,1,\cdots,n-1$$

证：偶 $\|f(x)-p(x)\|_\infty = \Delta(f,p)$

$\|f(-x)-p(-x)\|_\infty = \Delta(f,p)$

$\Rightarrow \|f(x)-p(-x)\|_\infty = \Delta(f,p)$

$\Rightarrow p(-x)$ 最佳一致逼近

$\Rightarrow p(-x)=p(x)$

奇 $\|f(x)-p(x)\|_\infty = \Delta(f,p)$

$\|f(-x)-p(-x)\|_\infty = \Delta(f,p)$

$\|f(x)+p(-x)\|_\infty = \Delta(f,p)$

$\Rightarrow -p(-x)$ 最佳一致逼近

$\Rightarrow p(x) = -p(-x)$

证：$p_{k+1}(x) = c\cdot\prod_{j=0}^{k}(x-x_j)$

$$(l_i(x), l_k(x)) = \int_a^b \rho(x) \frac{p_{k+1}(x)}{c_1(x-x_i)} \cdot \frac{p_{k+1}(x)}{c_2(x-x_k)}\,dx \quad (i\neq k)$$

$$= \frac{1}{c_1 c_2}\int_a^b \rho(x)\, p_{k+1}(x)\, s_{k-1}(x)\,dx$$

其中 $s_{k-1}(x)$ 为最高项为 x^{k-1} 的多项式

即 $s_{k-1}(x) \in \mathrm{span}\{p_{k-1}(x),\cdots,p_0(x)\}$

$$\therefore \frac{1}{c_1 c_2}\int_a^b \rho(x)\, p_{k+1}(x)\, s_{k-1}(x)\,dx = 0$$

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